solutions

September 9, 2026

1 Chapter 2: Statistical Learning Homework

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2, 4, 8, 10

2. Explain whether each scenario is a classification or regression problem, and indicate whether we are most interested in inference or prediction. Finally, provide `n` and `p`.

   a. We collect a set of data on the top 500 firms in the US. For each firm, we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.

   Ans: This is a *regression* problem since we're dealing with salary, which is a numerical value. We're most interested in *inference* becaure we're analyzing the relationship between salary and other factors.

   `n` = 500, `p` = 3.

   b. We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

   Ans: This is a *classification* problem since we determine succcess or failure. We're most interested in *prediction* since we're predicing an outcome $\hat{Y}$ for given inputs X.

   `n` = 20, `p` = 13.

   c. We are interested in predicting the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence we collect weekly data for all of 2012. For each week we record the % change in the USD/Euro, the % change in the US market, the % change in the British market, and the % change in the German market.

   Ans: This is a *regression* problem since we're predicting numerical data. We're most interested in *inference* because we're doing our prediction with world markets in mind.

   `n` = 52, `p` = 3.

4. You will now think of some real-life applications for statistical learning.

   a. Describe three real-life applications in which classification might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction? Explain your answer.

1. Spam email detection. Response would be binary spam/not spam. Predictors could be frequencies of specific words or word patterns like repeating exclamation marks or sender domain reputation. The goal would be prediction- to accurately classify if new email is spam.
2. Hospital readmission risk. Response would be binary – patient will be readmitted within 30 days or will not. Predictors could be age, primary diagnosis, number of prior hospital visits, length of initial stay, lab test results, and medication adherence history. The goal would be prediction – to accurately flag high-risk patients at discharge so the hospital can schedule extra follow-up care.
3. Credit card fraud detection. Response would be binary – fraudulent transaction or legitimate transaction. Predictors could be transaction amount, time of day, merchant category, geographic location of the purchase, and the user's typical spending patterns over recent weeks. The goal would be prediction – to instantly flag suspicious transactions in real time to block them before they are approved.

b. Describe three real-life applications in which regression might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction? Explain your answer.

1. Real estate price estimation. Response would be continuous – the sale price of a house in dollars. Predictors could be square footage, number of bedrooms and bathrooms, age of the property, lot size, neighborhood crime rate, and distance to the nearest school. The goal would be prediction – to provide an accurate market value for any new listing that appears on a real estate platform.
2. Forecasting daily electricity demand. Response would be continuous – megawatt-hours of electricity consumed on a given day. Predictors could be forecasted high temperature, day of the week, a holiday indicator, and historical demand from the same day in the previous year. The goal would be prediction – to anticipate tomorrow's load so the power grid operator can schedule generation and avoid blackouts.
3. Pharmaceutical drug dosage optimization. Response would be continuous – the reduction in a patient's blood pressure measured in mmHg after taking a new drug. Predictors could be dosage level, patient age, weight, baseline blood pressure, kidney function, and specific genetic markers. The goal would be inference – to estimate the precise dose–response curve and understand how different patient subgroups respond, in order to determine the optimal safe dosage for each group.

c. Describe three real-life applications in which cluster analysis might be useful.

1. Customer market segmentation. There is no predefined response variable. The data could include each customer's annual spending on various product categories (groceries, electronics, clothing, home goods), frequency of store visits, and average cart size. The goal would be inference – to discover natural groupings of customers (such as budget-conscious families or tech enthusiasts) so that management can understand the market landscape and design targeted marketing campaigns.
2. Document topic discovery. There is no predefined response variable. The data could include term-frequency–inverse-document-frequency vectors for all words appearing in a collection of news articles. The goal would be inference – to uncover latent thematic

structures (such as politics, sports, or technology) without manual labeling, allowing a news aggregator to organise its content and recommend similar articles.

3. Anomaly detection in network traffic. There is no predefined response variable. The data could include for each network connection its duration, number of bytes transferred, protocol type, number of failed login attempts, and source/destination port numbers. The goal would be inference – to characterise what normal traffic looks like so that any observation falling far outside the established clusters can be flagged as a potential cyberattack, even though we never have pre-labeled "attack" examples during training.

8. This exercise relates to the `College` data set.

[2]:
```python
# a
import pandas as pd

college = pd.read_csv('../datasets/College.csv')
```

[3]:
```python
# b
college
```

[3]:
```
                         Unnamed: 0 Private   Apps  Accept  Enroll  Top10perc  \
0      Abilene Christian University     Yes   1660    1232     721         23
1                Adelphi University     Yes   2186    1924     512         16
2                    Adrian College     Yes   1428    1097     336         22
3               Agnes Scott College     Yes    417     349     137         60
4         Alaska Pacific University     Yes    193     146      55         16
..                              …     …    …     …     …        …
772          Worcester State College      No   2197    1515     543          4
773               Xavier University     Yes   1959    1805     695         24
774  Xavier University of Louisiana     Yes   2097    1915     695         34
775                 Yale University     Yes  10705    2453    1317         95
776    York College of Pennsylvania     Yes   2989    1855     691         28

     Top25perc  F.Undergrad  P.Undergrad  Outstate  Room.Board  Books  \
0           52         2885          537      7440        3300    450
1           29         2683         1227     12280        6450    750
2           50         1036           99     11250        3750    400
3           89          510           63     12960        5450    450
4           44          249          869      7560        4120    800
..         …          …          …       …         …    …
772         26         3089         2029      6797        3900    500
773         47         2849         1107     11520        4960    600
774         61         2793          166      6900        4200    617
775         99         5217           83     19840        6510    630
776         63         2988         1726      4990        3560    500

     Personal  PhD  Terminal  S.F.Ratio  perc.alumni  Expend  Grad.Rate
0        2200   70        78       18.1           12    7041         60
```

```
1        1500   29         30        12.2          16   10527          56
2        1165   53         66        12.9          30    8735          54
3         875   92         97         7.7          37   19016          59
4        1500   76         72        11.9           2   10922          15
..          …    …          …         …          …     …      …
772      1200   60         60        21.0          14    4469          40
773      1250   73         75        13.3          31    9189          83
774       781   67         75        14.4          20    8323          49
775      2115   96         96         5.8          49   40386          99
776      1250   75         75        18.1          28    4509          99

[777 rows x 19 columns]
```

[4]:

```python
# b
college2 = pd.read_csv('../datasets/College.csv', index_col=0)
college3 = college.rename({'Unnamed: 0': 'College'}, axis=1)
college3 = college3.set_index('College')
```

[5]:

```python
# b
college = college3
```

[6]:

```python
# c
college.describe
```

[6]:

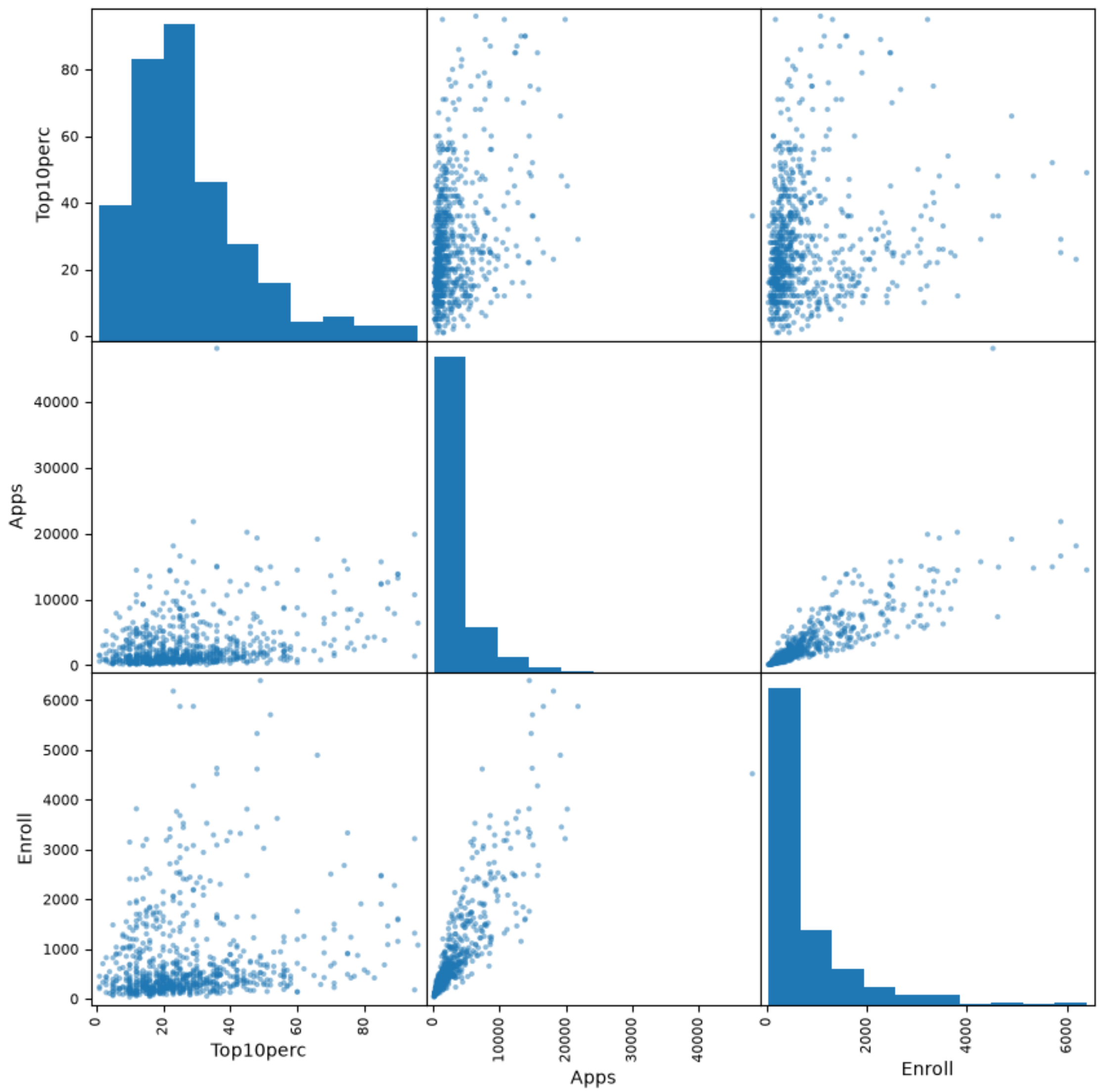
```
<bound method NDFrame.describe of                                 Private   Apps
Accept  Enroll  Top10perc  \
College
Abilene Christian University       Yes   1660    1232     721         23
Adelphi University                 Yes   2186    1924     512         16
Adrian College                     Yes   1428    1097     336         22
Agnes Scott College                Yes    417     349     137         60
Alaska Pacific University          Yes    193     146      55         16
…                                 …    …      …      …      …
Worcester State College             No   2197    1515     543          4
Xavier University                  Yes   1959    1805     695         24
Xavier University of Louisiana     Yes   2097    1915     695         34
Yale University                    Yes  10705    2453    1317         95
York College of Pennsylvania       Yes   2989    1855     691         28

                                Top25perc  F.Undergrad  P.Undergrad  Outstate  \
College
Abilene Christian University           52         2885          537      7440
Adelphi University                     29         2683         1227     12280
Adrian College                         50         1036           99     11250
Agnes Scott College                    89          510           63     12960
Alaska Pacific University              44          249          869      7560
…                                    …          …           …       …
```

```
Worcester State College                 26          3089          2029       6797
Xavier University                       47          2849          1107      11520
Xavier University of Louisiana          61          2793           166       6900
Yale University                         99          5217            83      19840
York College of Pennsylvania            63          2988          1726       4990

                                Room.Board  Books  Personal  PhD  Terminal  \
College
Abilene Christian University          3300    450      2200   70        78
Adelphi University                    6450    750      1500   29        30
Adrian College                        3750    400      1165   53        66
Agnes Scott College                   5450    450       875   92        97
Alaska Pacific University             4120    800      1500   76        72
…                                        …      …         …    …         …
Worcester State College               3900    500      1200   60        60
Xavier University                     4960    600      1250   73        75
Xavier University of Louisiana        4200    617       781   67        75
Yale University                       6510    630      2115   96        96
York College of Pennsylvania          3560    500      1250   75        75

                                S.F.Ratio  perc.alumni  Expend  Grad.Rate
College
Abilene Christian University         18.1           12    7041         60
Adelphi University                   12.2           16   10527         56
Adrian College                       12.9           30    8735         54
Agnes Scott College                   7.7           37   19016         59
Alaska Pacific University            11.9            2   10922         15
…                                       …            …       …          …
Worcester State College              21.0           14    4469         40
Xavier University                    13.3           31    9189         83
Xavier University of Louisiana       14.4           20    8323         49
Yale University                       5.8           49   40386         99
York College of Pennsylvania         18.1           28    4509         99

[777 rows x 18 columns]>
```

[7]:
```python
# d
pd.plotting.scatter_matrix(
    college[['Top10perc', 'Apps', 'Enroll']],
    figsize=(10, 10)
)
```

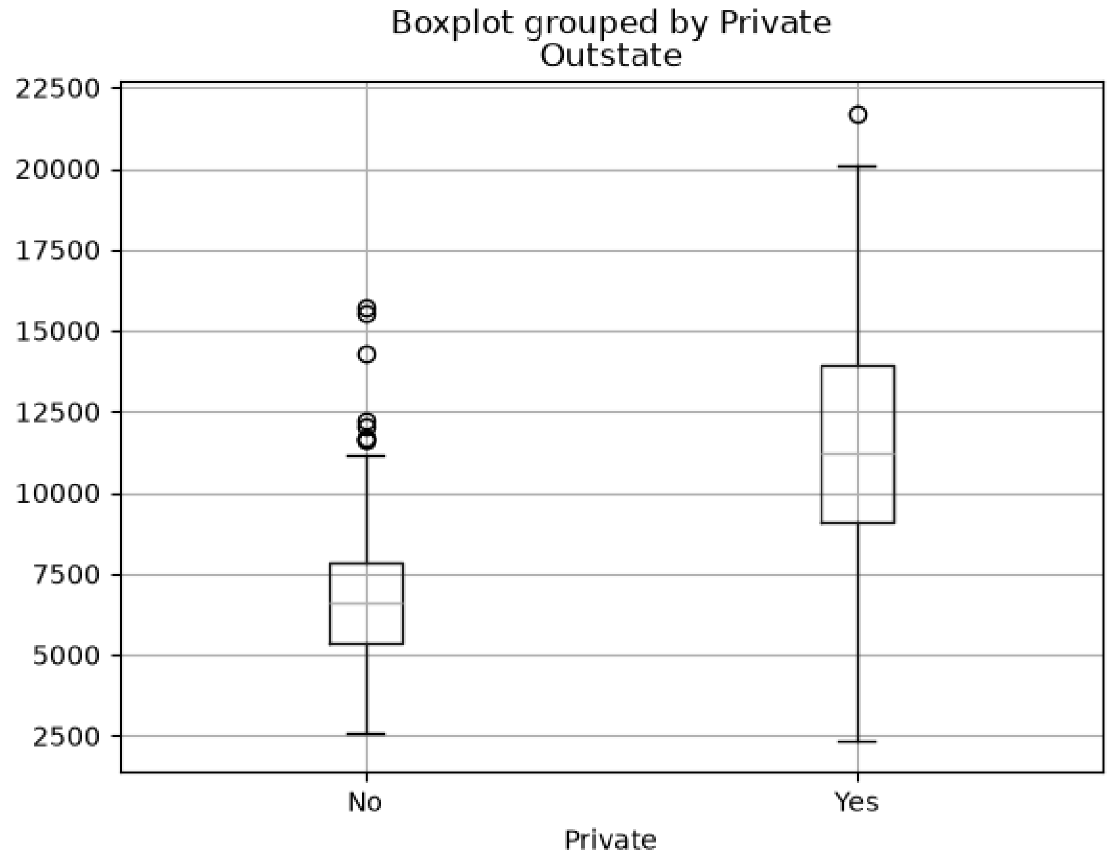
[7]:
```
array([[<Axes: xlabel='Top10perc', ylabel='Top10perc'>,
        <Axes: xlabel='Apps', ylabel='Top10perc'>,
        <Axes: xlabel='Enroll', ylabel='Top10perc'>],
       [<Axes: xlabel='Top10perc', ylabel='Apps'>,
        <Axes: xlabel='Apps', ylabel='Apps'>,
```

```
 <Axes: xlabel='Enroll', ylabel='Apps'>],
[<Axes: xlabel='Top10perc', ylabel='Enroll'>,
 <Axes: xlabel='Apps', ylabel='Enroll'>,
 <Axes: xlabel='Enroll', ylabel='Enroll'>]], dtype=object)
```



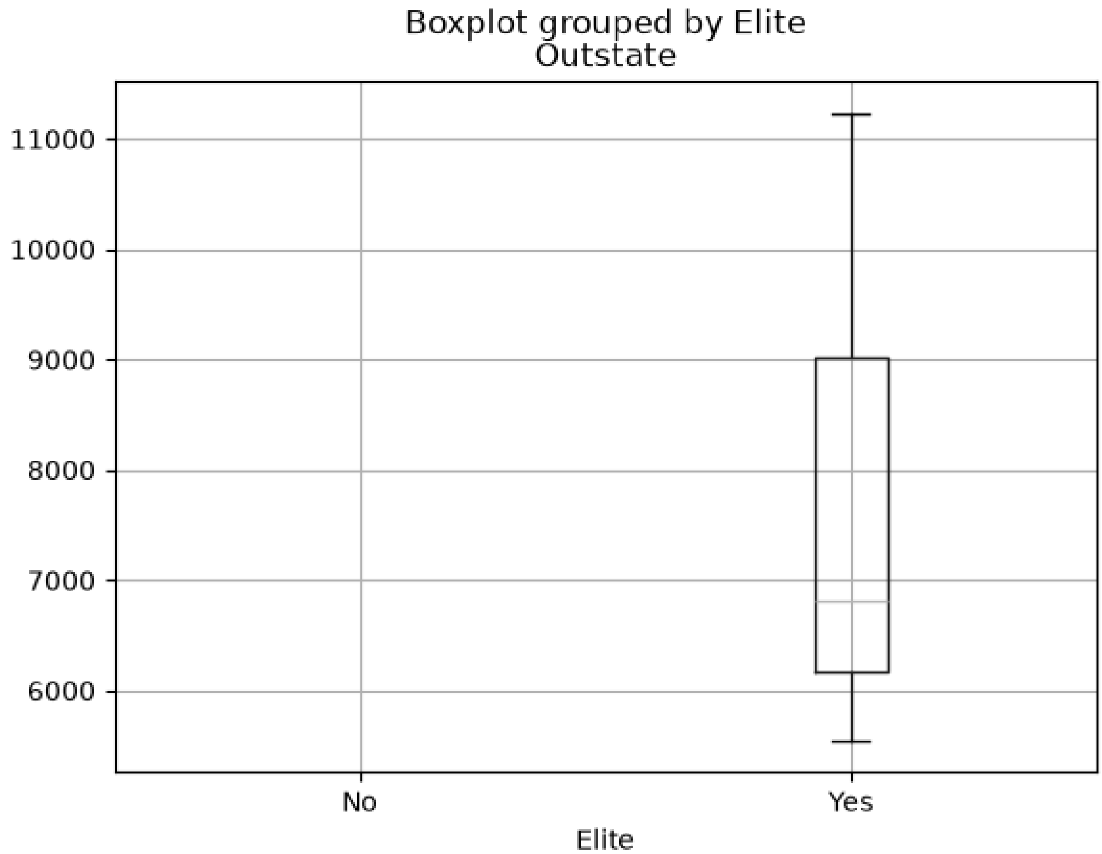
[10]:
```
# d
college.boxplot(column='Outstate', by='Private')
```

[10]:
```
<Axes: title={'center': 'Outstate'}, xlabel='Private'>
```

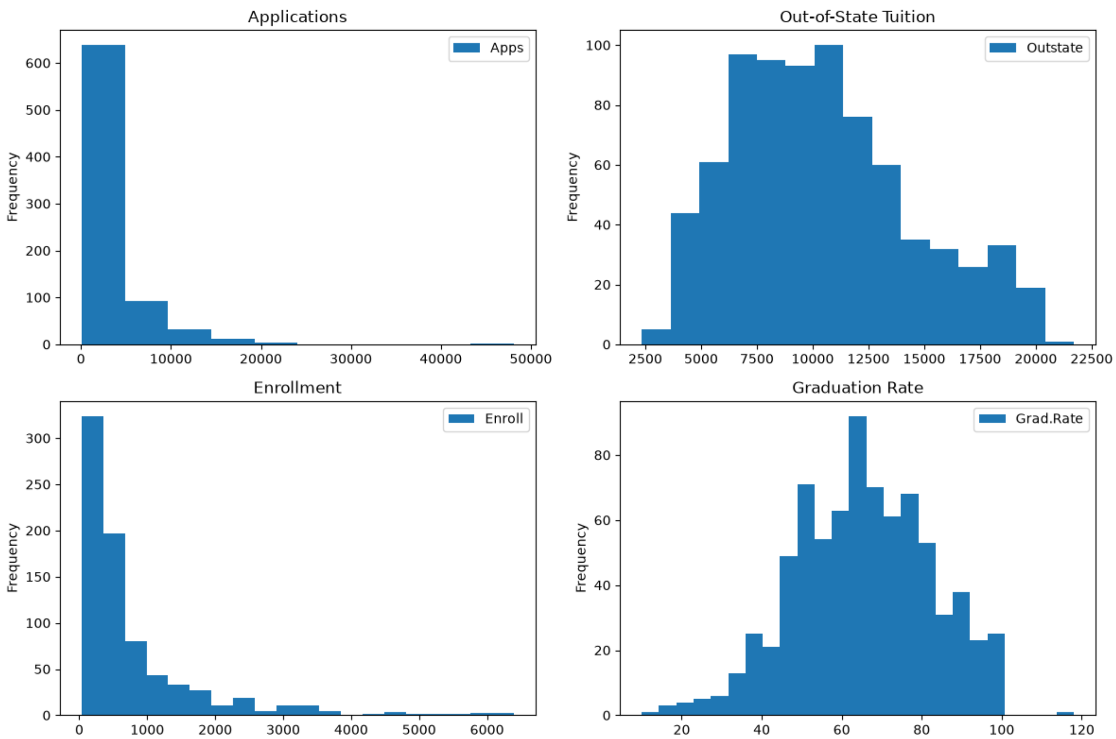


[11]:
```
# f
college['Elite'] = pd.cut(college['Top10perc'], [0, 0.5, 1], labels=['No',␣
  ↪'Yes'])
college['Elite'].value_counts()
college.boxplot(column='Outstate', by='Elite')
```

[11]: <Axes: title={'center': 'Outstate'}, xlabel='Elite'>



[12]:
```
# g
import matplotlib.pyplot as plt

fig, axes = plt.subplots(2, 2, figsize=(12, 8))

college.plot.hist(column='Apps', bins=10, ax=axes[0, 0], title='Applications')
college.plot.hist(column='Outstate', bins=15, ax=axes[0, 1],␣
↪title='Out-of-State Tuition')
college.plot.hist(column='Enroll', bins=20, ax=axes[1, 0], title='Enrollment')
college.plot.hist(column='Grad.Rate', bins=25, ax=axes[1, 1], title='Graduation␣
↪Rate')

plt.tight_layout()
plt.show()
```

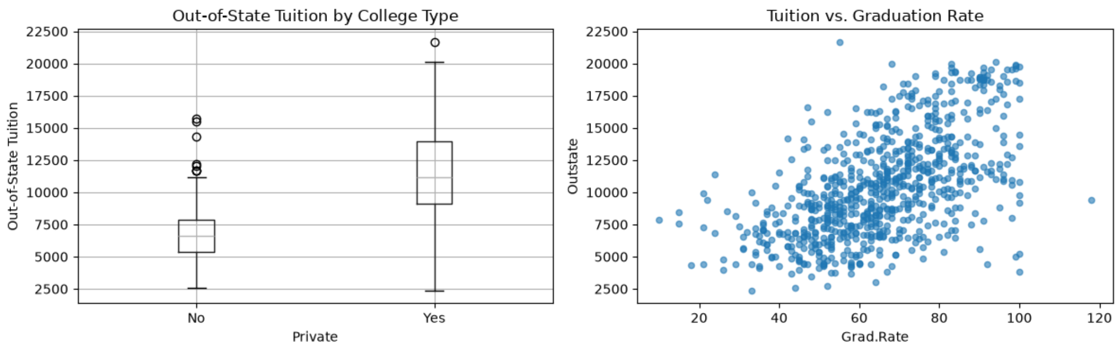


[13]:
```python
# h
# Basic structure and missing values
print(f"Observations: {college.shape[0]}")
print(f"Variables: {college.shape[1]}")
print("\nMissing values:")
print(college.isna().sum()[lambda x: x > 0])

# Compare private and public colleges
group_summary = college.groupby('Private', observed=False)[
    ['Apps', 'Enroll', 'Outstate', 'Grad.Rate']
].mean().round(2)

print("\nMean values by college type:")
print(group_summary)

# Strongest correlations with out-of-state tuition
correlations = (
    college.select_dtypes('number')
    .corr()['Outstate']
    .drop('Outstate')
    .abs()
    .sort_values(ascending=False)
)
```

```
print("\nVariables most strongly associated with Outstate tuition:")
print(correlations.head(5))

# Additional visual exploration
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

college.boxplot(column='Outstate', by='Private', ax=axes[0])
axes[0].set_title('Out-of-State Tuition by College Type')
axes[0].set_xlabel('Private')
axes[0].set_ylabel('Out-of-State Tuition')

college.plot.scatter(
    x='Grad.Rate',
    y='Outstate',
    ax=axes[1],
    alpha=0.6
)
axes[1].set_title('Tuition vs. Graduation Rate')

plt.suptitle('')
plt.tight_layout()
plt.show()

print(
    "\nSummary: Private colleges generally have higher out-of-state tuition "
    "and graduation rates than public colleges. Tuition also shows a positive "
    "association with graduation rate, although the relationship is not perfect.
 ↪"
)
```

```
Observations: 777
Variables: 19

Missing values:
Elite    774
dtype: int64

Mean values by college type:
            Apps   Enroll  Outstate  Grad.Rate
Private
No       5729.92  1640.87   6813.41      56.04
Yes      1977.93   456.95  11801.69      69.00

Variables most strongly associated with Outstate tuition:
Expend         0.672779
Room.Board     0.654256
Grad.Rate      0.571290
```

```
perc.alumni    0.566262
Top10perc      0.562331
Name: Outstate, dtype: float64
```



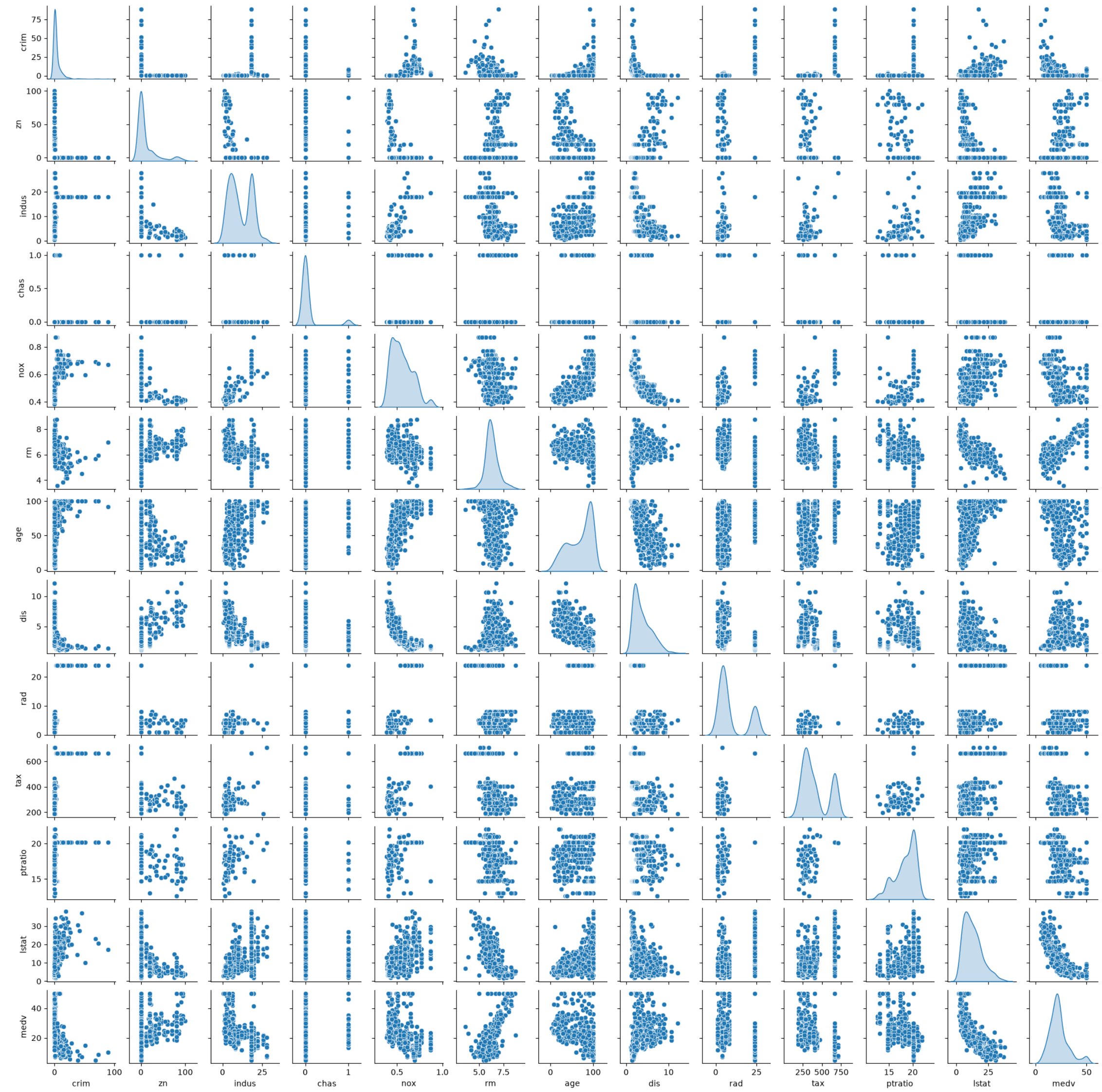

```
Summary: Private colleges generally have higher out-of-state tuition and
graduation rates than public colleges. Tuition also shows a positive association
with graduation rate, although the relationship is not perfect.
```

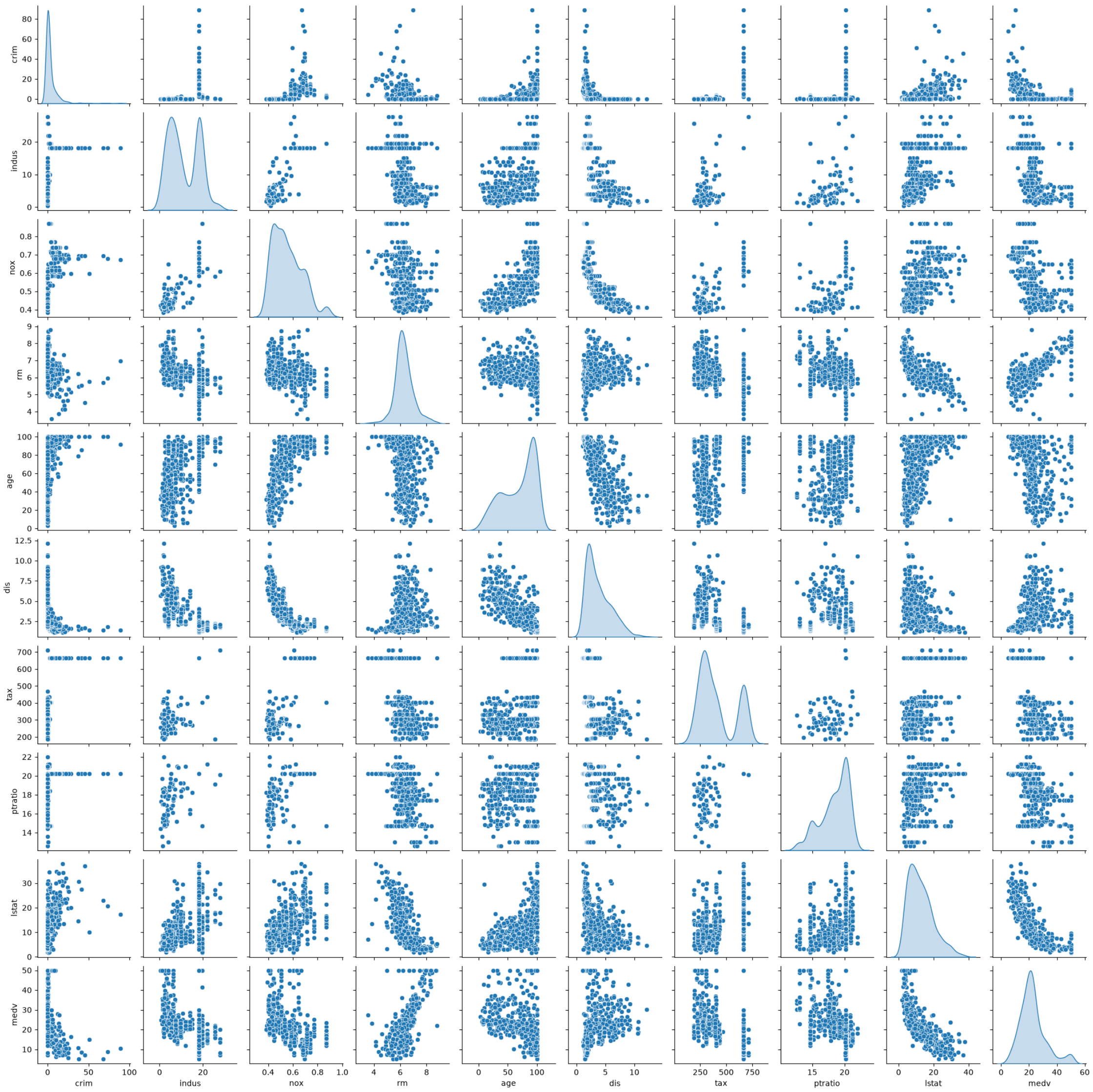
10. This exercise involves the Boston housing data set.

[16]:
```python
# a
from ISLP import load_data

boston = load_data("Boston")
```

b. Thre are 506 rows and 13 columns. The rows represent one suburban area of Boston, and the columns represent different attributes of those areas, such as crime rate, average number of rooms, and median value of owner-occupied homes. They are 12 predictor variables and one response variable.

[17]:
```python
# c

import seaborn as sns
import matplotlib.pyplot as plt

# Option 1 - all 13 variables (will be large - 13×13 plots)
sns.pairplot(boston, diag_kind='kde', height=1.5)
plt.show()

# Option 2 - a focused subset (recommended for readability)
cols = ['crim', 'indus', 'nox', 'rm', 'age', 'dis', 'tax', 'ptratio', 'lstat',␣
 ↪'medv']
sns.pairplot(boston[cols], diag_kind='kde', height=2)
plt.show()
```


crim
zn
indus
chas
nox
rm
age
dis
rad
tax
ptratio
lstat
medv



From the scatterplots:

- Crime (`crim`) rises with `indus`, `nox`, `age`, `rad`, `tax`, `lstat` – i.e., industrial, polluted, older, high-tax, lower-status areas have more crime.
- Crime falls with `zn`, `rm`, `dis`, `medv` – more residential zoning, larger homes, greater distance from employment, and higher home values associate with lower crime.
- Home value (`medv`) climbs strongly with `rm` (larger rooms) and drops sharply with `lstat`, `age`, and `nox` –.
- Outliers are visible: `crim` ranges up to 89 (most are under 4), and `medv` ranges from 5 to 50, so scatterplots show some extreme points that skew the relationships.

d. Yes, crime rate (`crim`) is positively associated with `indus`, `nox`, `age`, `rad`, `tax`, and `lstat` (more industry, pollution, older housing, highway access, taxes, and lower-status population drive crime up), and negatively associated with `zn`, `rm`, `dis`, and `medv` (more residential zoning, larger rooms, greater distance from employment centres, and higher home values drive crime down).

e. Yes, some suburbs have extreme crime rates (max 88.98 vs. mean 3.61), tax rates (max 711, with the 75th percentile already at 666), and pupil-teacher ratios (max 22.0, 75th percentile 20.2). Overall ranges span from 0.006 to 88.98 for crime, 187 to 711 for tax, and 12.6 to 22.0 for pupil-teacher ratio, indicating that a few outliers skew the upper tails heavily.

f. Exactly 35 suburbs in the dataset border the Charles River since `chas` is a binary indicator and its sum is 35.

g. The median pupil-teacher ratio among all towns is 19.05.

h. The suburb with the lowest home value (`medv` = 5.0) is row 398, which has extreme values: `crim` = 38.35, `age` = 100, `rad` = 24, `tax` = 666, `ptratio` = 20.2, and `lstat` = 30.59 – all at or near their maxima – while `rm` = 5.453 and `dis` = 1.49 are well below average. This suburb is pretty much the worst in every socio-economic and environmental dimension, explaining its rock bottom home price.

i. 64 suburbs average more than 7 rooms per dwelling, and 13 average more than 8. Among those 13, most have very high home values and low crime with one exception, index 364, which has the highest average rooms (8.78) yet only a `medv` of 21.9, due to high crime (3.47), heavy industry (18.1), high taxes (666), and poor student teacher ratio (20.2), showing that even large homes cannot fully compensate for severe neighbourhood disadvantages.